GAMM and GAMLSS examples

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Packages

```
rm(list = ls())
# data and plotting
library(dplyr)
library(GGally)
# models
library(mgcv)
library(gamlss)
# for some of the optimization algorithms
set.seed(123)
```

Data

Select monthly average water quality measurements in Chesapeake Bay for station CB5.1[1]:

- DO: concentration of dissolved oxygen;
- CHLA: concentration of chlorophyll-a;
- SALINITY: water salinity;
- WTEMP: water temperature;
- TSS: concentration of total suspended solids;
- TN: concentration of total nitrogen;
- TP: concentration of total phosphorus.

```
D <- read.csv("./data/CBmonthly.csv") %>%
  dplyr::filter(is.element(Station, c("CB5.1"))) %>%
  dplyr::select(Station, Year, Month, DO, CHLA, SALINITY, WTEMP, TSS, TN, TP) %>%
  mutate(Time = Year + Month/12 - 1/24,
         Station = factor(Station))
str(D)
```

```
## 'data.frame':    384 obs. of  11 variables:
##  $ Station : Factor w/ 1 level "CB5.1": 1 1 1 1 1 1 1 1 1 1 ...
##  $ Year    : int  1985 1985 1985 1985 1985 1985 1985 1985 1985 1985 ...
##  $ Month   : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ DO      : num  11.35 12.5 11.93 12.11 8.53 ...
##  $ CHLA    : num  13.09 10.86 16.21 20.36 6.99 ...
##  $ SALINITY: num  14.7 15.6 13.9 12.9 13.6 ...
##  $ WTEMP   : num  4.9 0.85 7.38 13.96 17.65 ...
##  $ TSS     : num  4.5 3.25 6.59 8.25 8.55 ...
##  $ TN      : num  0.736 0.713 0.722 0.905 0.689 ...
```

[1] https://www.chesapeakebay.net/documents/3676/map_of_mainstem_and_tributary_monitoring_stations.pdf

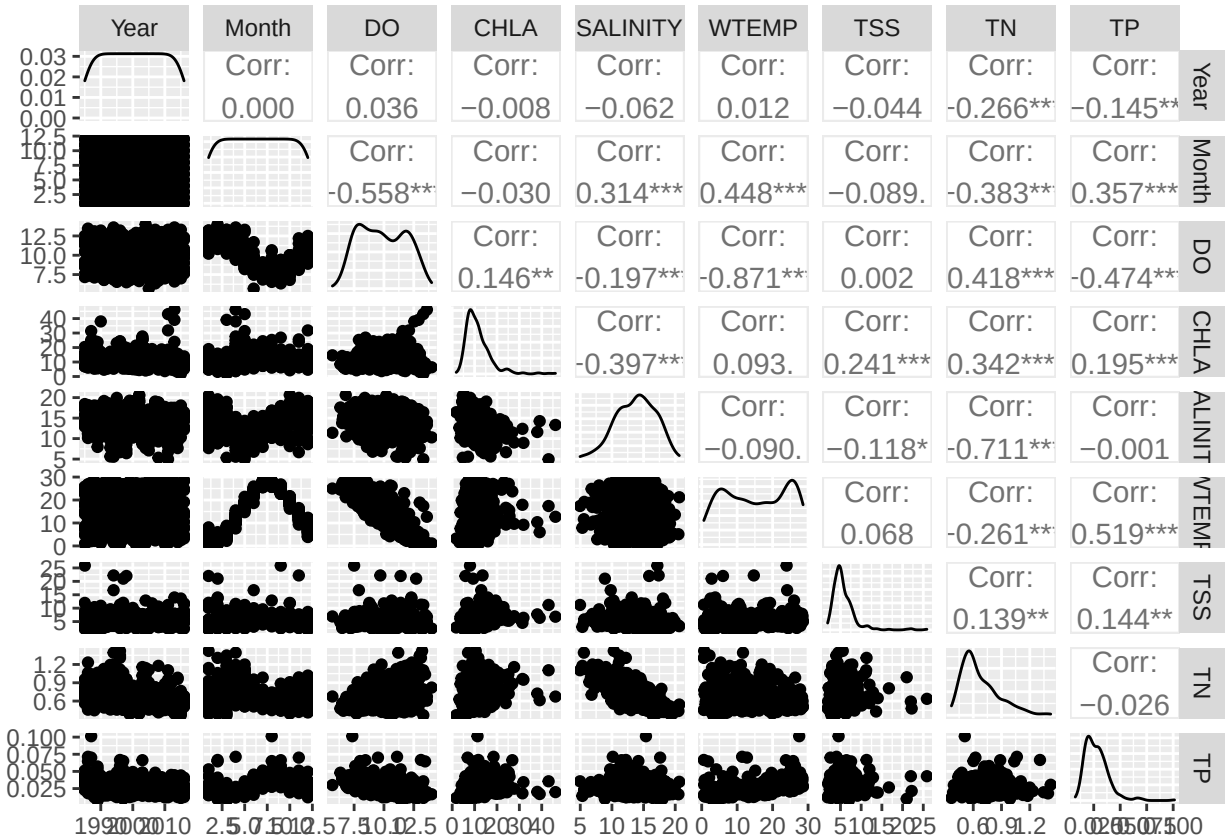
```
##  $ TP      : num  0.0238 0.0375 0.0331 0.0236 0.0229 ...
##  $ Time    : num  1985 1985 1985 1985 1985 ...
```

```
psych::describe(D)
```

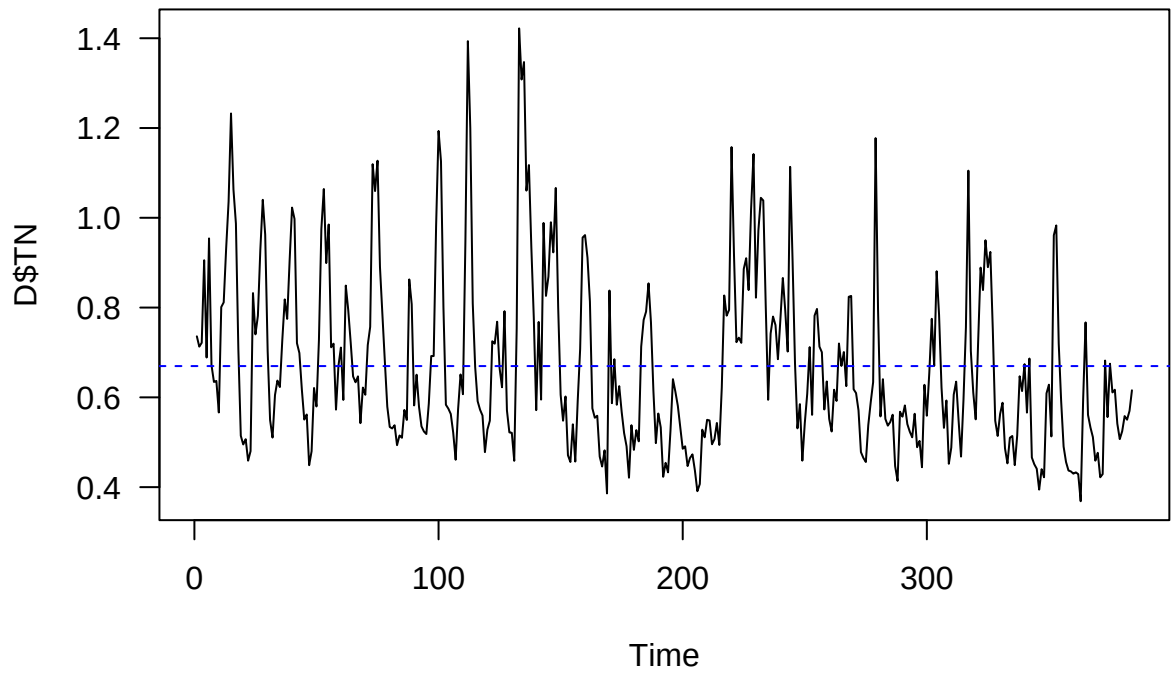
```
##          vars   n    mean   sd  median trimmed   mad     min     max range
## Station*    1 384    1.00 0.00    1.00    1.00  0.00    1.00    1.00  0.00
## Year        2 384 2000.50 9.25 2000.50 2000.50 11.86 1985.00 2016.00 31.00
## Month       3 384    6.50 3.46    6.50    6.50  4.45    1.00   12.00 11.00
## DO          4 384    9.88 1.92    9.74    9.85  2.58    5.68   14.00  8.32
## CHLA        5 384   11.44 5.70   10.18   10.65  4.10    1.78   46.14 44.36
## SALINITY    6 384   13.76 3.08   13.93   13.85  3.29    4.96   20.64 15.68
## WTEMP       7 384   15.38 8.70   15.65   15.50 12.68    0.60   28.60 28.00
## TSS         8 384    5.66 2.76    5.06    5.28  1.83    1.90   25.88 23.98
## TN          9 384    0.67 0.20    0.61    0.65  0.16    0.37    1.42  1.05
## TP         10 384    0.03 0.01    0.03    0.03  0.01    0.01    0.10  0.09
## Time       11 384 2001.00 9.25 2001.00 2001.00 11.86 1985.04 2016.96 31.92
##           skew kurtosis   se
## Station*   NaN      NaN 0.00
## Year      0.00    -1.21 0.47
## Month     0.00    -1.23 0.18
## DO        0.10    -1.17 0.10
## CHLA      2.23     8.14 0.29
## SALINITY -0.27    -0.41 0.16
## WTEMP    -0.05    -1.41 0.44
## TSS       3.11    15.62 0.14
## TN        1.13     1.04 0.01
## TP        1.67     5.99 0.00
## Time      0.00    -1.21 0.47
```

Plot data

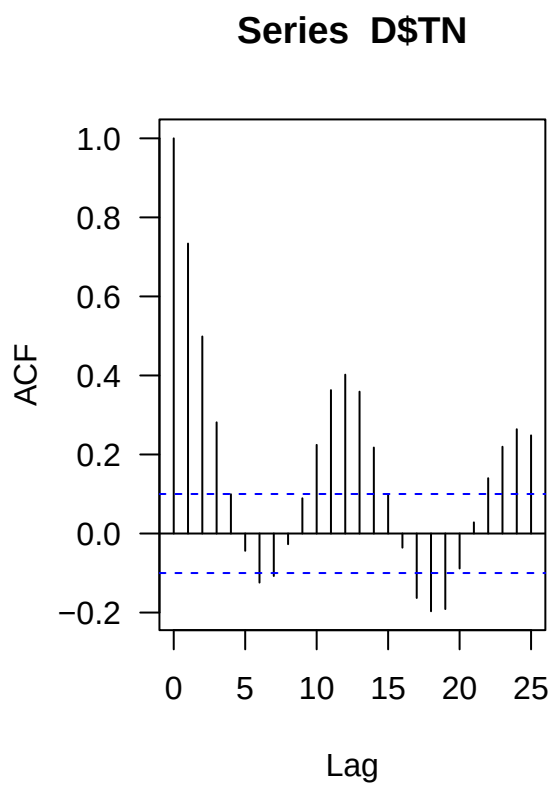
```
D %>%
  dplyr::select(-Station, -Time) %>%
  ggpairs()
```

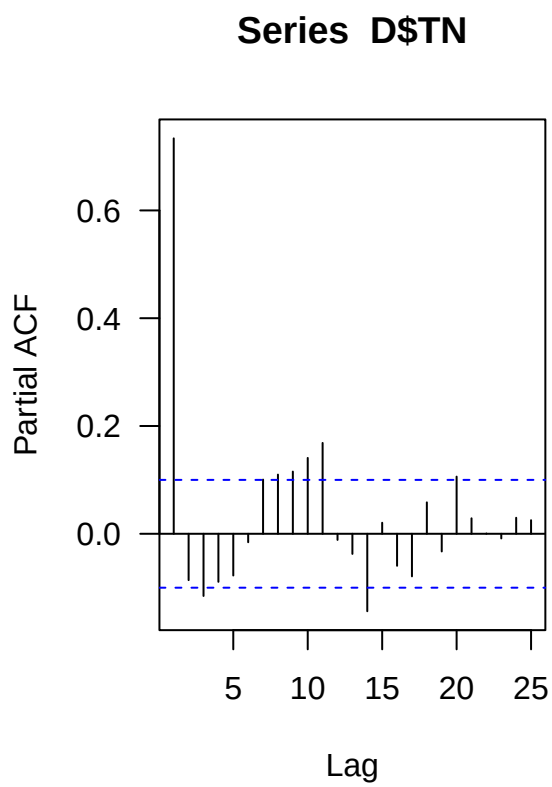



```
plot.ts(D$TN, las = 1)
abline(h = mean(D$TN), lty = 2, col = "blue")
```




```
par(mfrow = c(1, 2))
acf(D$TN, las = 1)
pacf(D$TN, las = 1)
```





Generalized additive model (GAM)

Fit the model

$$TN_t = \beta_0 + f(Year_t) + f(Month_t) + f(Salinity_t) + e_t,$$

where $e_t \sim N(0, \sigma^2)$.

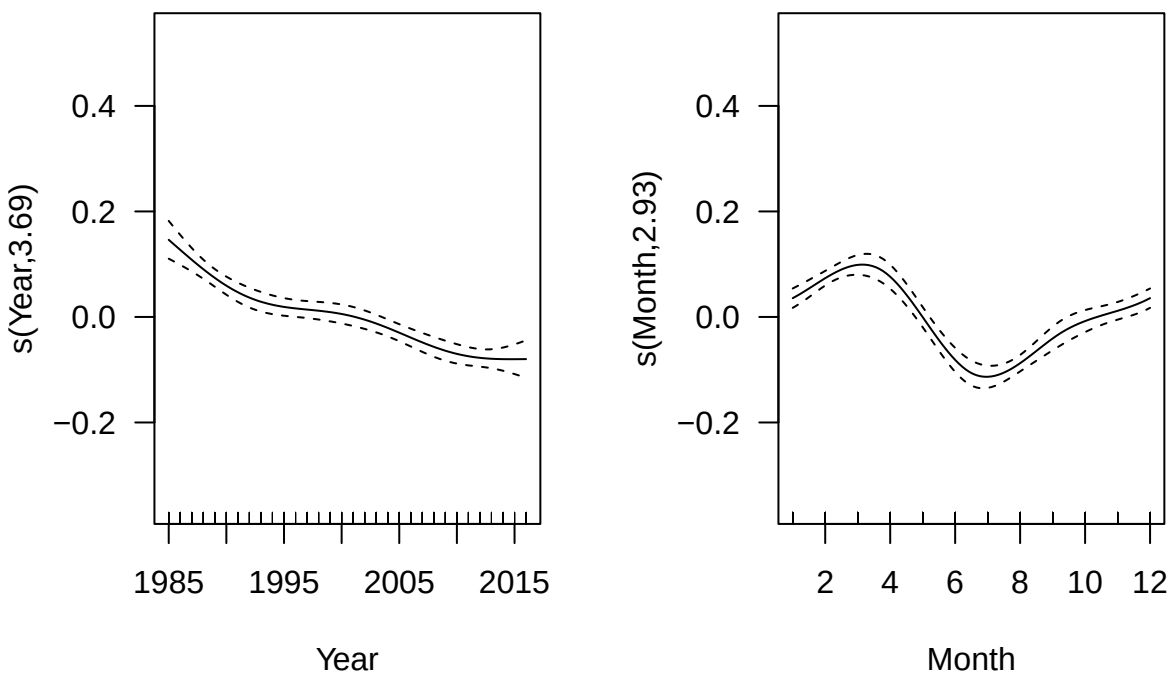
```
m_gam <- gam(TN ~ s(Year, k = 5) + s(Month, bs = "cc", k = 5)
             + s(SALINITY, k = 5)
             ,family = gaussian
             ,data = D)
summary(m_gam)
```

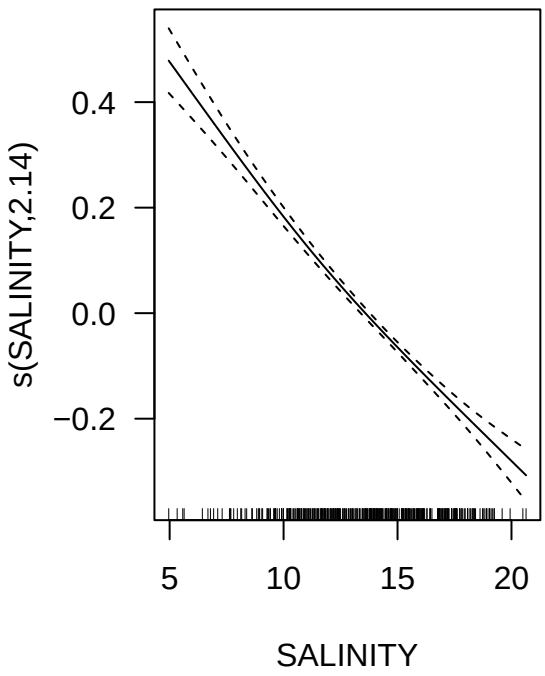
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## TN ~ s(Year, k = 5) + s(Month, bs = "cc", k = 5) + s(SALINITY,
##     k = 5)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.669719   0.005196   128.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Approximate significance of smooth terms:
##                edf Ref.df      F p-value
## s(Year)      3.689  3.941  37.20  <2e-16 ***
## s(Month)     2.931  3.000  53.71  <2e-16 ***
## s(SALINITY) 2.138  2.654 202.59  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.729   Deviance explained = 73.5%
## GCV = 0.010637  Scale est. = 0.010366  n = 384
```

Visualize the smooth terms. For identifiability, sum of the values of each curve, at the observed covariate values, must be zero.

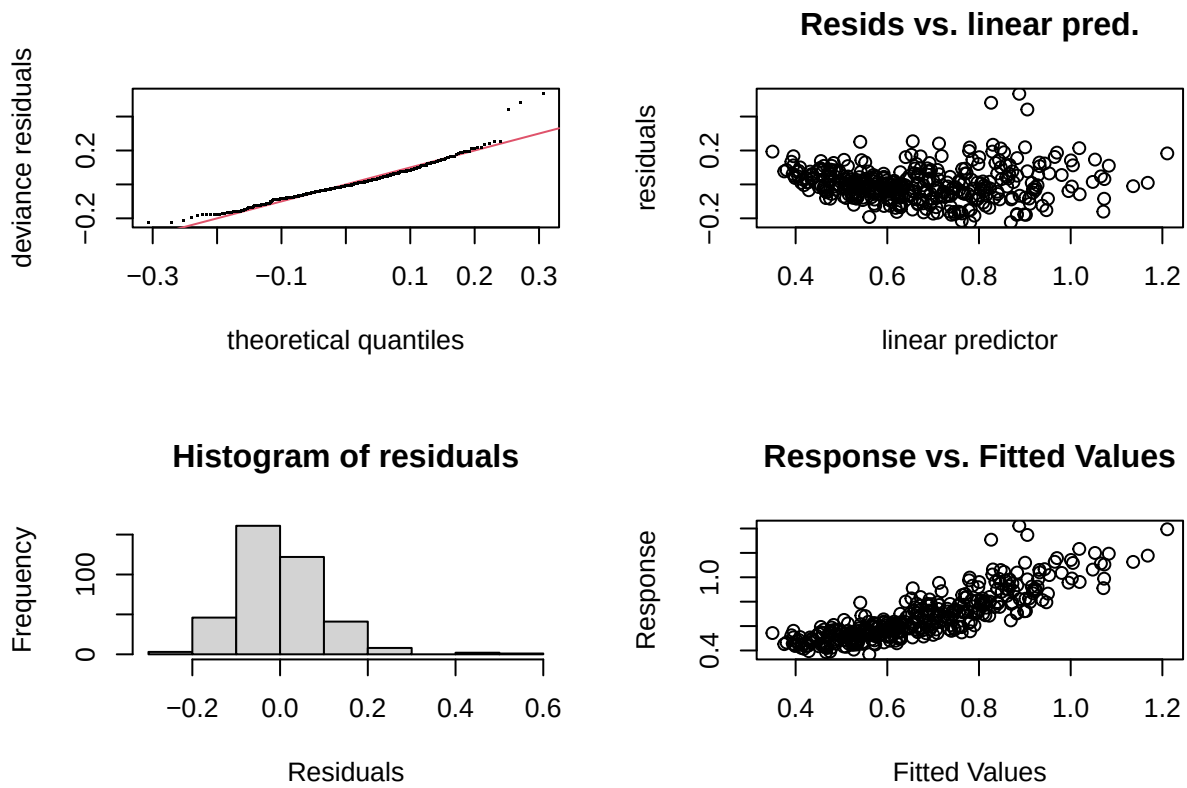
```r
par(mfrow = c(1, 2))
plot(m_gam, las = 1)
```





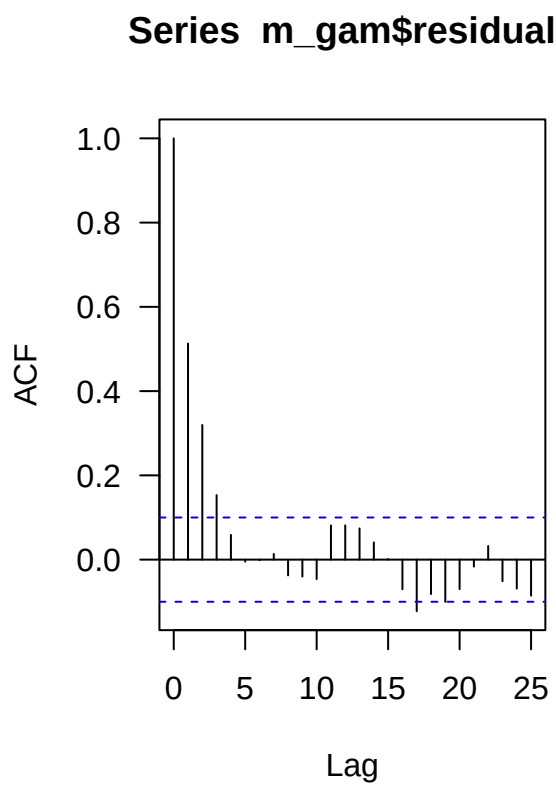
Diagnostics

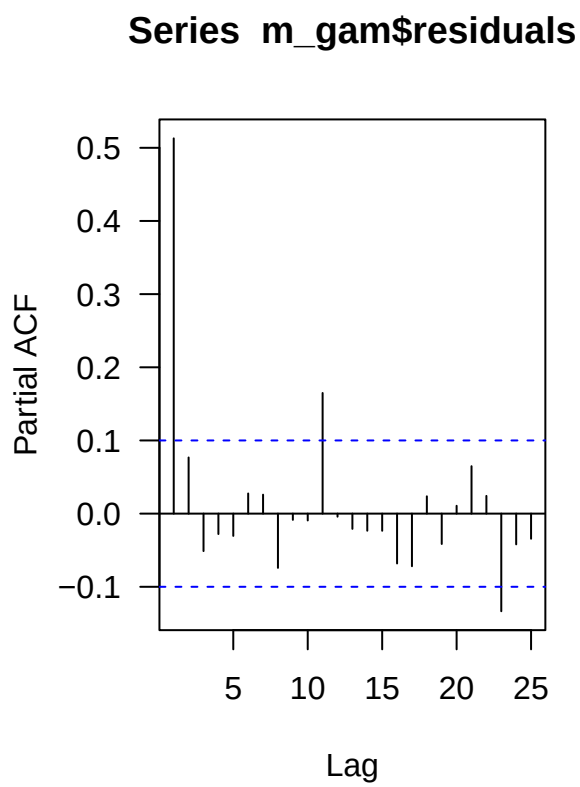
```
gam.check(m_gam)
```




```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 8 iterations.
## The RMS GCV score gradient at convergence was 4.374481e-07 .
## The Hessian was positive definite.
## Model rank =  12 / 12
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                k'  edf k-index p-value
## s(Year)      4.00 3.69    0.49  <2e-16 ***
## s(Month)     3.00 2.93    0.92   0.085 .
## s(SALINITY) 4.00 2.14    1.11   0.985
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
par(mfrow = c(1, 2))
acf(m_gam$residuals, las = 1)
pacf(m_gam$residuals, las = 1)
```





Generalized additive mixed model (GAMM)

Fit the model

$$TN_t = \beta_0 + f(Year_t) + f(Month_t) + f(Salinity_t) + e_t,$$

where $e_t \sim AR(1)$.

```
m_gamm <- gamm(TN ~ s(Year, k = 5) + s(Month, bs = "cc", k = 5)
               + s(SALINITY, k = 5)
               ,correlation = corARMA(p = 1)
               ,family = gaussian
               ,data = D)
summary(m_gamm)
```

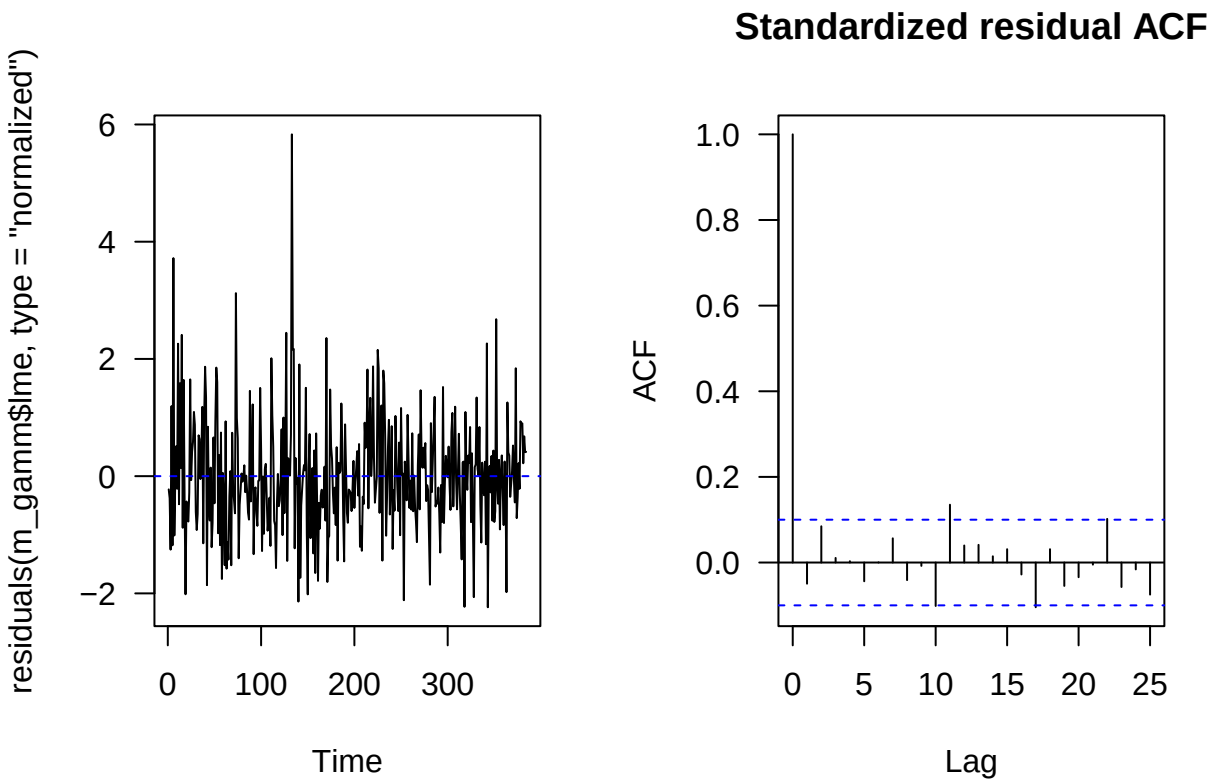
```
##     Length Class Mode
## lme 18     lme   list
## gam 31     gam   list
```

```
summary(m_gamm$lme)
```

```
## Linear mixed-effects model fit by maximum likelihood
##   Data: strip.offset(mf)
##         AIC       BIC   logLik
##   -757.0545 -725.4494 386.5273
##
## Random effects:
##  Formula: ~Xr - 1 | g
```

```
##  Structure: pdIdnot
##                  Xr1          Xr2          Xr3
## StdDev: 1.021847e-05 1.021847e-05 1.021847e-05
##
##  Formula: ~Xr.0 - 1 | g.0 %in% g
##  Structure: pdIdnot
##              Xr.01      Xr.02      Xr.03
## StdDev: 0.06077295 0.06077295 0.06077295
##
##  Formula: ~Xr.1 - 1 | g.1 %in% g.0 %in% g
##  Structure: pdIdnot
##             Xr.11     Xr.12     Xr.13  Residual
## StdDev: 0.1666308 0.1666308 0.1666308 0.1027286
##
## Correlation Structure: AR(1)
##  Formula: ~1 | g/g.0/g.1
##  Parameter estimate(s):
##       Phi
## 0.5402587
## Fixed effects:  y ~ X - 1
##                      Value   Std.Error  DF   t-value p-value
## X(Intercept)     0.6699142 0.009604119 381  69.75280       0
## Xs(Year)Fx1     -0.0595191 0.009554507 381  -6.22943       0
## Xs(SALINITY)Fx1 -0.1832342 0.017540499 381 -10.44635       0
##  Correlation:
##                 X(Int) X(Y)F1
## Xs(Year)Fx1     0.000
## Xs(SALINITY)Fx1 0.003  0.018
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -2.1173355 -0.6479829 -0.1174745  0.5581450  5.2794708
##
## Number of Observations: 384
## Number of Groups:
##                   g          g.0 %in% g g.1 %in% g.0 %in% g
##                   1                   1                   1
```

```r
par(mfrow = c(1, 2))
plot.ts(residuals(m_gamm$lme, type = "normalized"), las = 1)
abline(h = 0, col = "blue", lty = 2)
acf(residuals(m_gamm$lme, type = "normalized"), las = 1, main = "Standardized residual ACF")
```



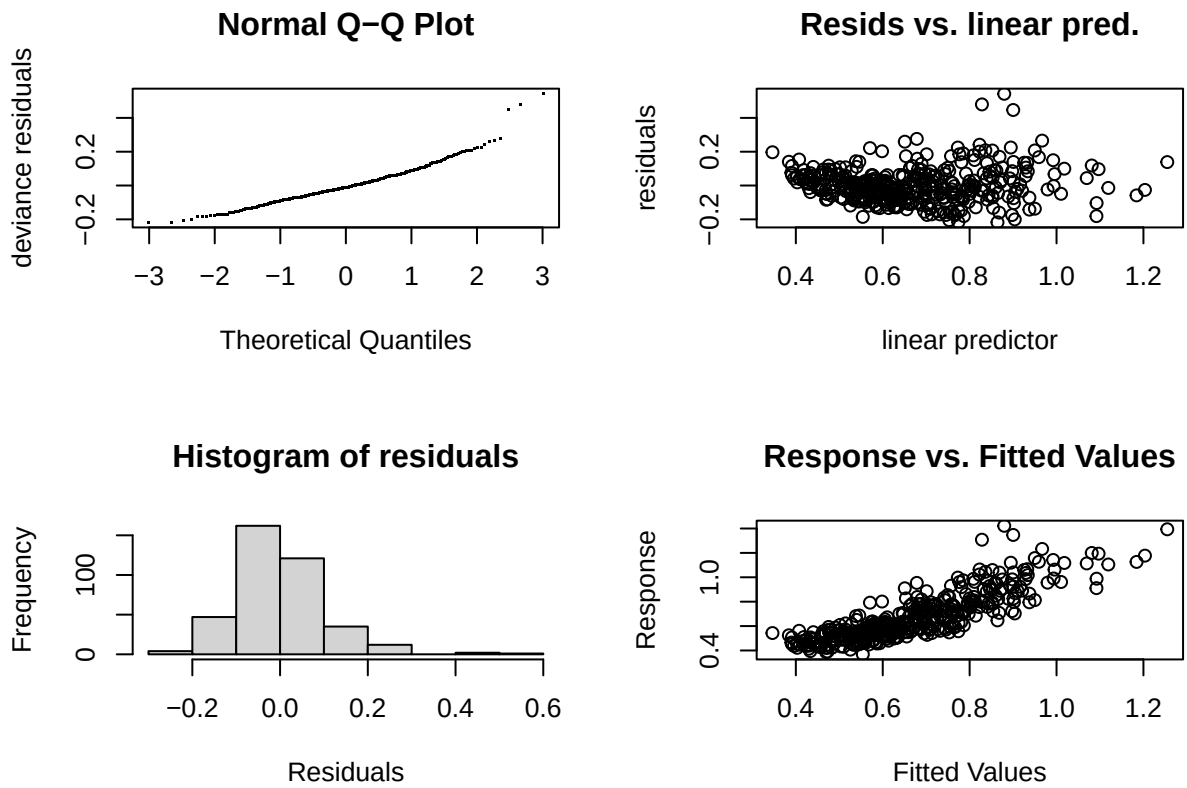

```
concurvity(m_gamm$gam)
```

```
##                   para     s(Year)   s(Month) s(SALINITY)
## worst    1.549454e-28 0.100259917 0.38754283   0.4122402
## observed 1.549454e-28 0.005928601 0.01991101   0.3374621
## estimate 1.549454e-28 0.016295767 0.15992826   0.3037763
```

```
concurvity(m_gamm$gam, full = FALSE)$estimate
```

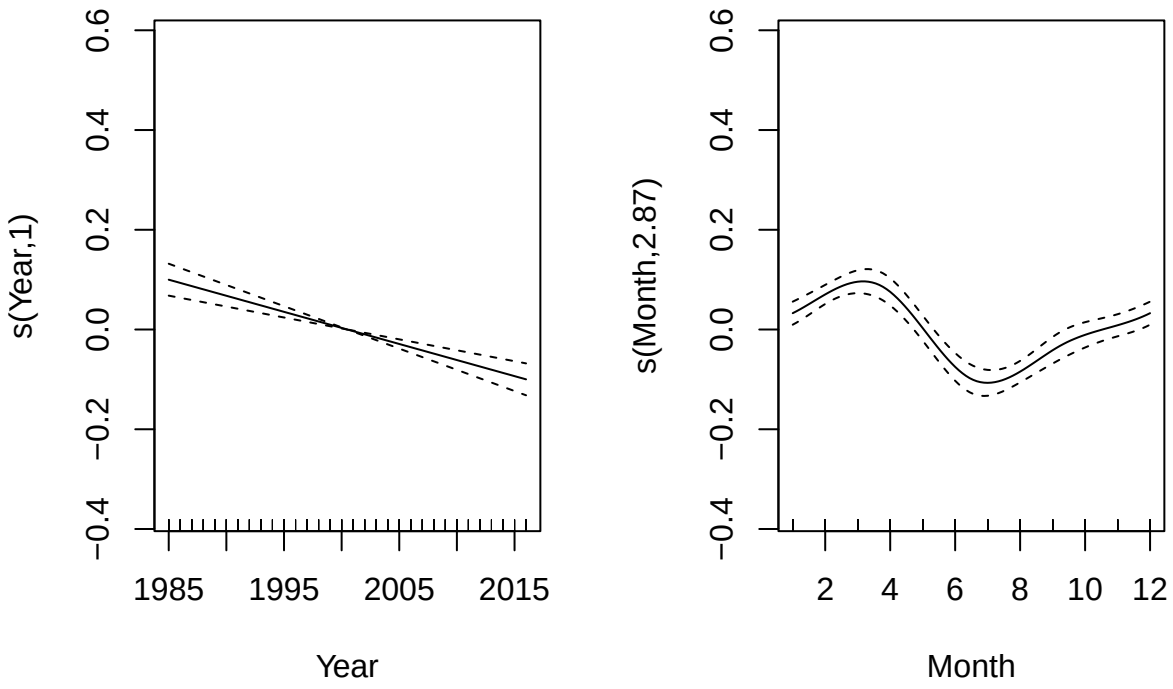
```
##                      para      s(Year)     s(Month)  s(SALINITY)
## para        1.000000e+00 1.460710e-30 1.350925e-32 4.203358e-31
## s(Year)     1.360128e-28 1.000000e+00 6.577389e-31 4.069689e-02
## s(Month)    3.677975e-32 7.272270e-33 1.000000e+00 2.630794e-01
## s(SALINITY) 1.964410e-29 1.401615e-02 1.537781e-01 1.000000e+00
```

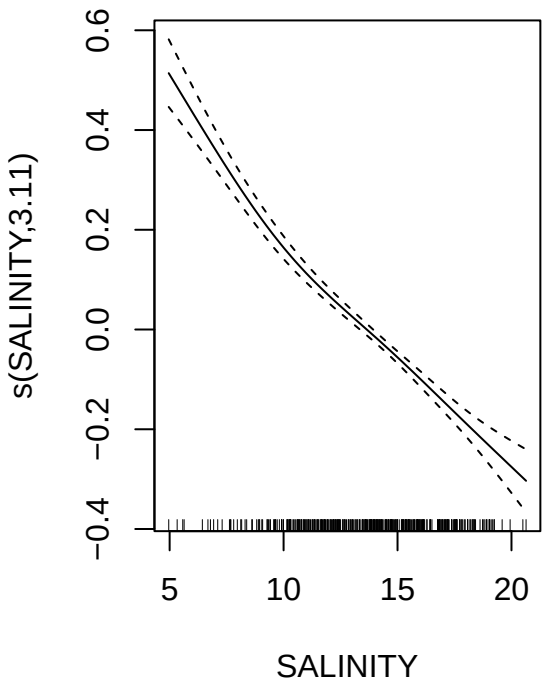
```
gam.check(m_gamm$gam)
```




```
##
## 'gamm' based fit - care required with interpretation.
## Checks based on working residuals may be misleading.
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                k'  edf k-index p-value
## s(Year)      4.00 1.00    0.46  <2e-16 ***
## s(Month)     3.00 2.87    0.89   0.005 **
## s(SALINITY) 4.00 3.11    1.09   0.935
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
par(mfrow = c(1, 2))
plot(m_gamm$gam)
```


s(Year,1)
0.6
0.4
0.2
0.0
−0.2
−0.4
1985
1995
2005
2015
Year
s(Month,2.87)
0.6
0.4
0.2
0.0
−0.2
−0.4
2
4
6
8
10
12
Month



Generalized additive model for location scale and shape (GAMLSS)

Fit the model

$$TN_t \sim SkewT(\mu_t, \sigma_t, \nu, \tau),$$
$$\mu_t = a_0 + f(Year_t) + f(Month_t) + f(Salinity_t) + f(TN_{t-1}),$$
$$\ln(\sigma_t) = b_0 + f(Year_t) + f(DO_t),$$
$$\nu = c_0,$$
$$\ln(\tau) = d_0.$$

```
D <- D %>%
  mutate(TN1 = dplyr::lag(TN, n = 1),
         TN2 = dplyr::lag(TN, n = 2)) %>%
  na.omit()
m_gamlss <- gamlss(TN ~ pb(Year) + pbc(Month) + pb(SALINITY) + pb(TN1) #+ TN2
                   # ,correlation = corARMA()
                   ,sigma.formula = ~pb(Year) + pb(DO)
                   ,family = ST1
                   ,control = gamlss.control(n.cyc = 100, c.crit = 0.9)
                   ,data = D)
```

```
## GAMLSS-RS iteration 1: Global Deviance = -843.6786
## GAMLSS-RS iteration 2: Global Deviance = -848.3079
## GAMLSS-RS iteration 3: Global Deviance = -852.6612
## GAMLSS-RS iteration 4: Global Deviance = -855.113
```

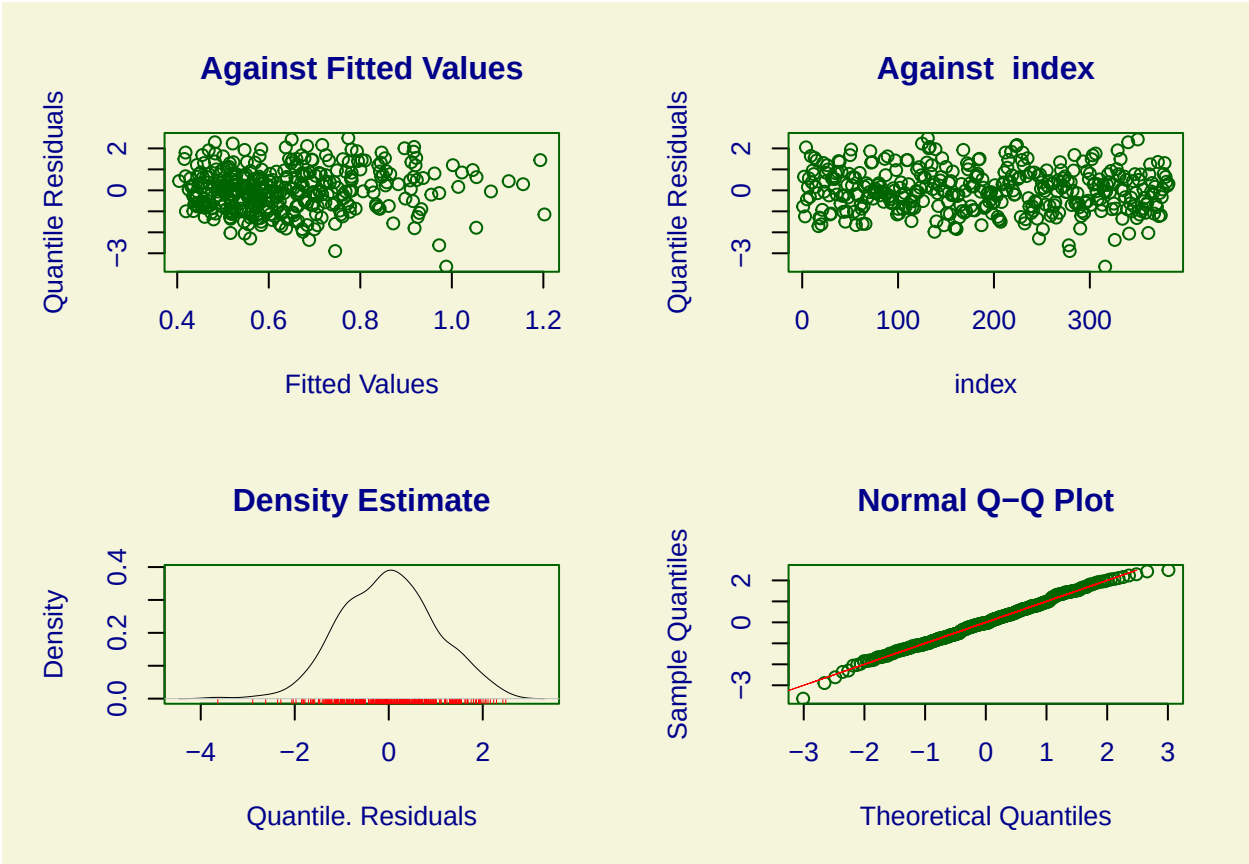
```
## GAMLSS-RS iteration 5: Global Deviance = -856.3218
## GAMLSS-RS iteration 6: Global Deviance = -857.1236
```

```
summary(m_gamlss)
```

```
## *******************************************************************
## Family:  c("ST1", "Skew t (Azzalini type 1)")
##
## Call:  gamlss(formula = TN ~ pb(Year) + pbc(Month) + pb(SALINITY) +
##     pb(TN1), sigma.formula = ~pb(Year) + pb(DO), family = ST1,
##     data = D, control = gamlss.control(n.cyc = 100, c.crit = 0.9))
##
## Fitting method: RS()
##
## -------------------------------------------------------------------
## Mu link function:  identity
## Mu Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.0527834  0.8180316   8.622  < 2e-16 ***
## pb(Year)     -0.0030838  0.0004015  -7.680  1.5e-13 ***
## pb(SALINITY) -0.0319588  0.0017313 -18.460  < 2e-16 ***
## pb(TN1)       0.2936712  0.0290238  10.118  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## -------------------------------------------------------------------
## Sigma link function:  log
## Sigma Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 41.731907  10.858560   3.843 0.000142 ***
## pb(Year)    -0.022950   0.005416  -4.238 2.84e-05 ***
## pb(DO)       0.138720   0.026652   5.205 3.19e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## -------------------------------------------------------------------
## Nu link function:  identity
## Nu Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.33747    0.06275   5.378 1.32e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## -------------------------------------------------------------------
## Tau link function:  log
## Tau Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.1927     0.1533    7.78  6.9e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## -------------------------------------------------------------------
## NOTE: Additive smoothing terms exist in the formulas:
##  i) Std. Error for smoothers are for the linear effect only.
## ii) Std. Error for the linear terms may not be reliable.
```

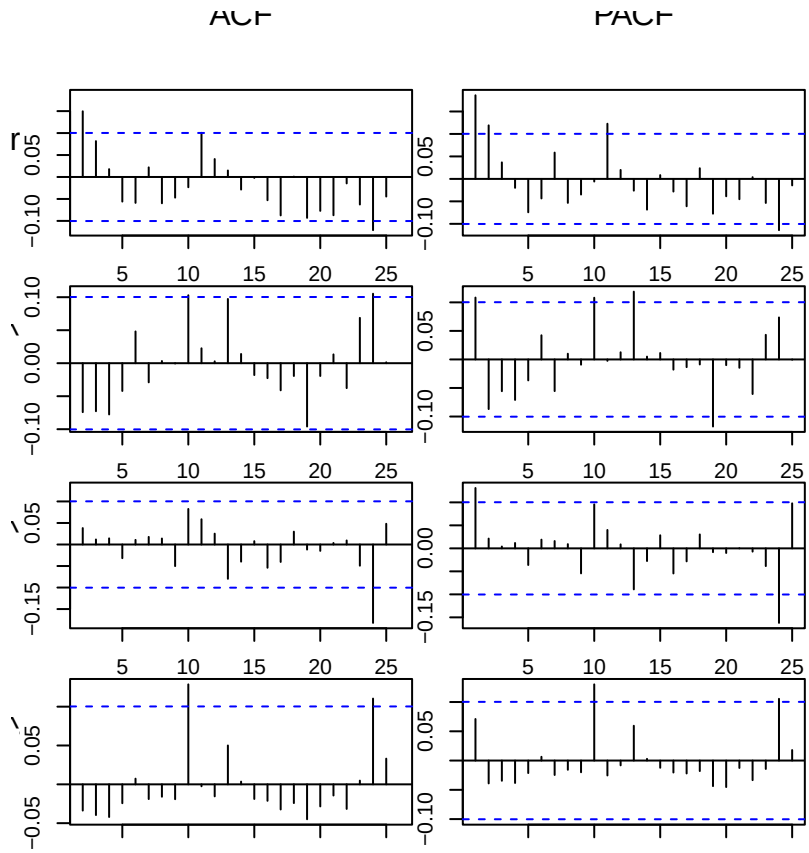
```
## -------------------------------------------------------------------
## No. of observations in the fit:  382
## Degrees of Freedom for the fit:  25.47459
##       Residual Deg. of Freedom:  356.5254
##                        at cycle:  6
##
## Global Deviance:     -857.1236
##             AIC:     -806.1744
##             SBC:     -705.6664
## ******************************************************************
```

```
plot(m_gamlss)
```

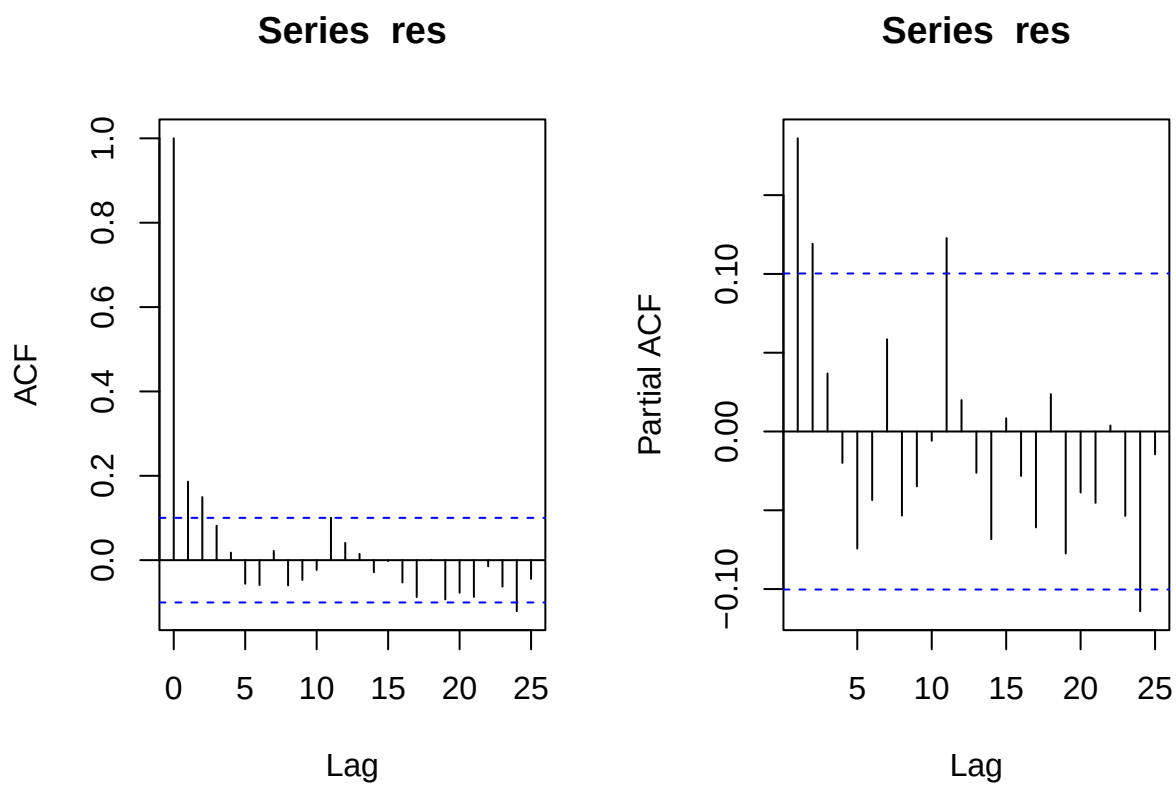



```
## ******************************************************************
##        Summary of the Quantile Residuals
##                             mean   =  0.001299784
##                         variance   =  1.002133
##                 coef. of skewness  =  -0.04775127
##                 coef. of kurtosis  =  2.981284
## Filliben correlation coefficient   =  0.9978631
## ******************************************************************
```

```
acfResid(m_gamlss)
```

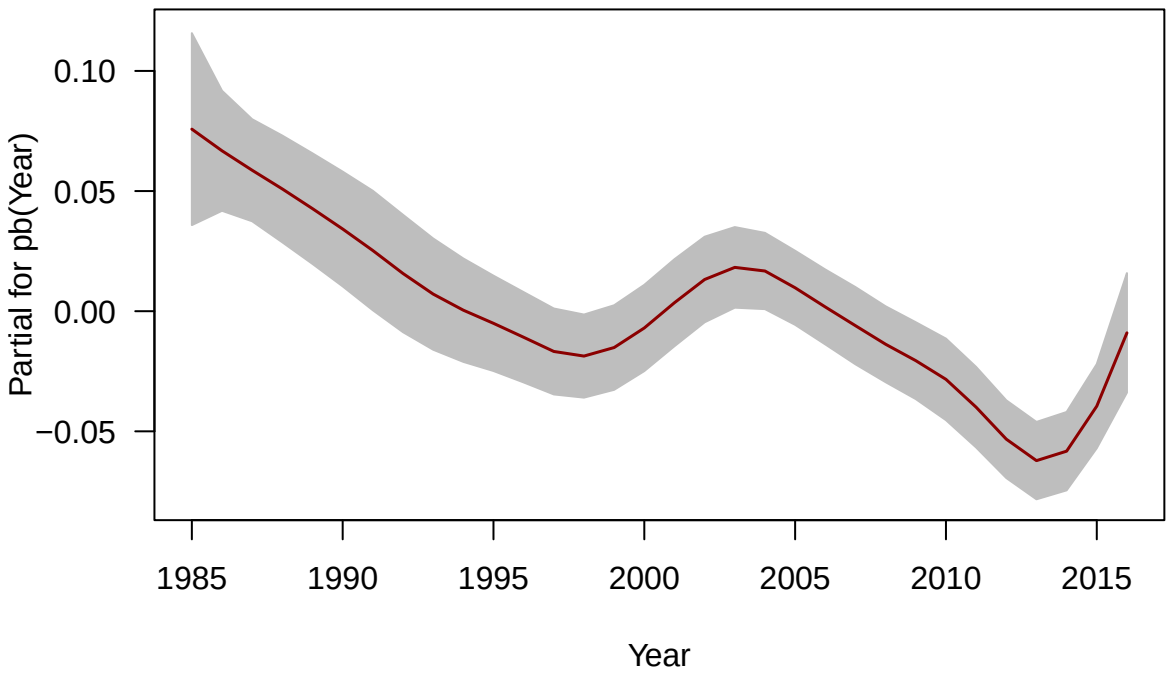


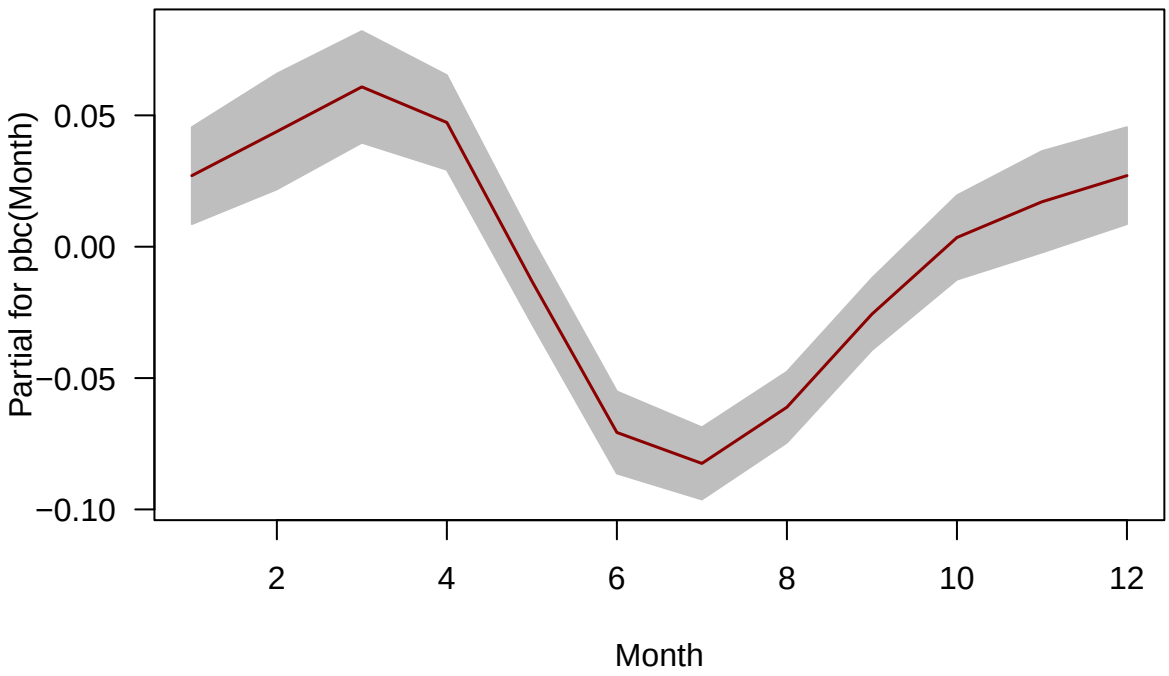

```
res <- residuals(m_gamlss, what = "z-scores")
par(mfrow = c(1, 2))
acf(res); pacf(res)
```

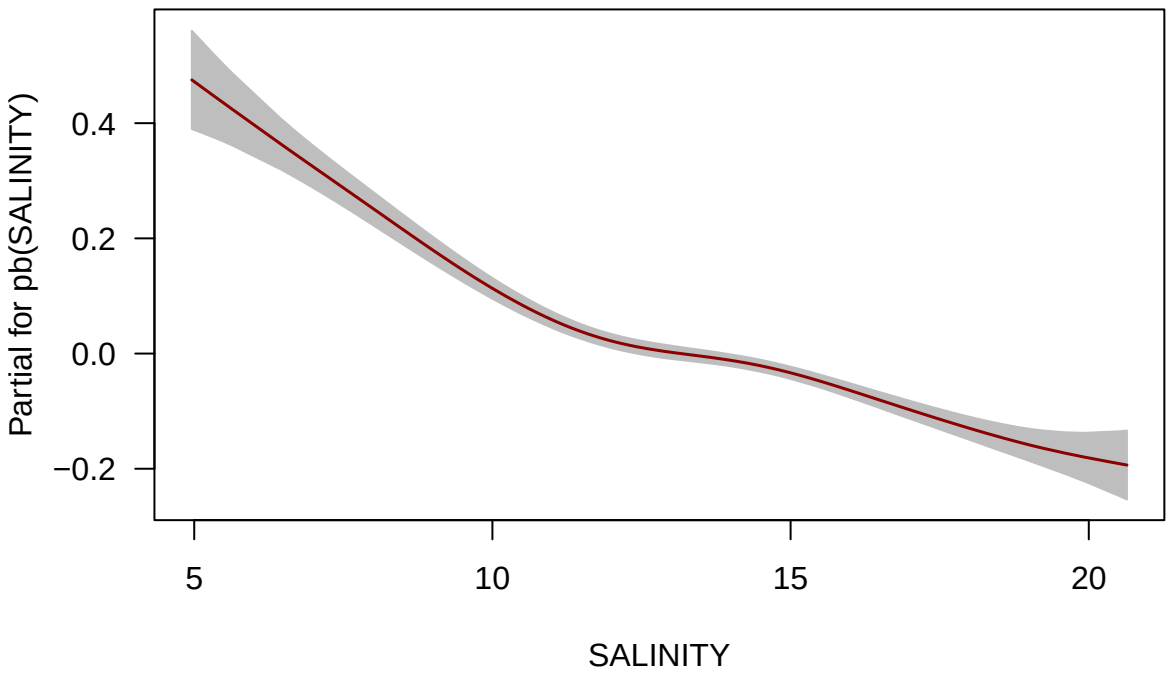


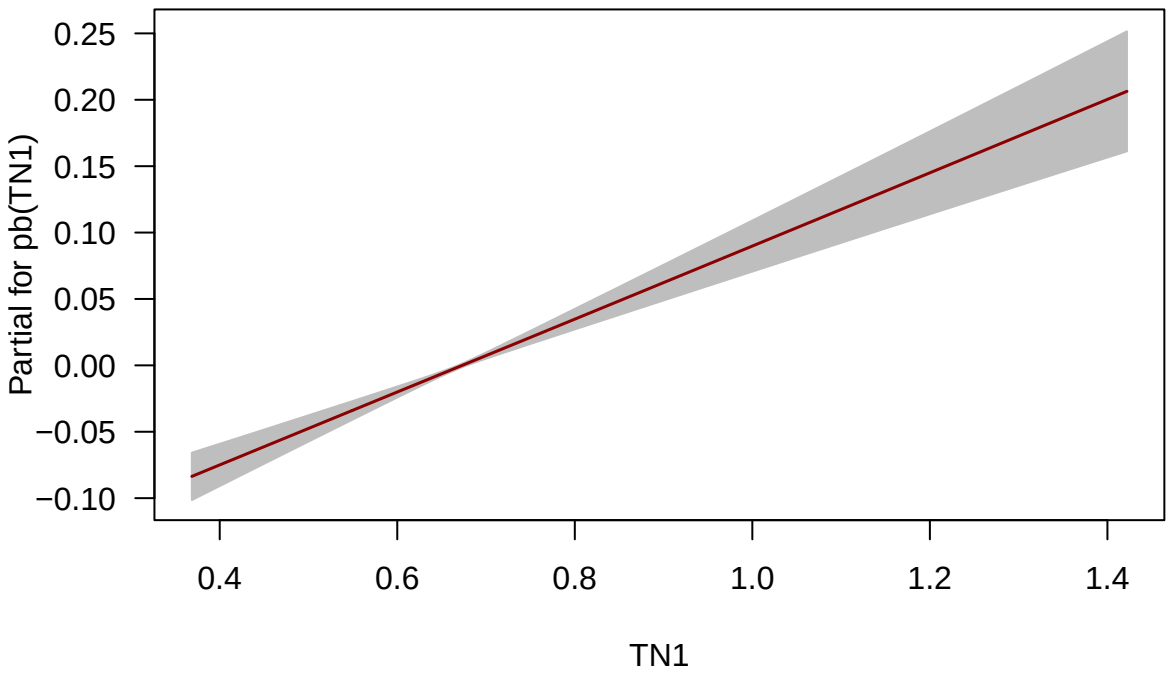
Smooth terms for the mean

```
term.plot(m_gamlss, las = 1, se = TRUE, ylim = "free") #, ylim = "common"
```


0.10
0.05
0.00
−0.05
Partial for pb(Year)
1985
1990
1995
2000
2005
2010
2015
Year

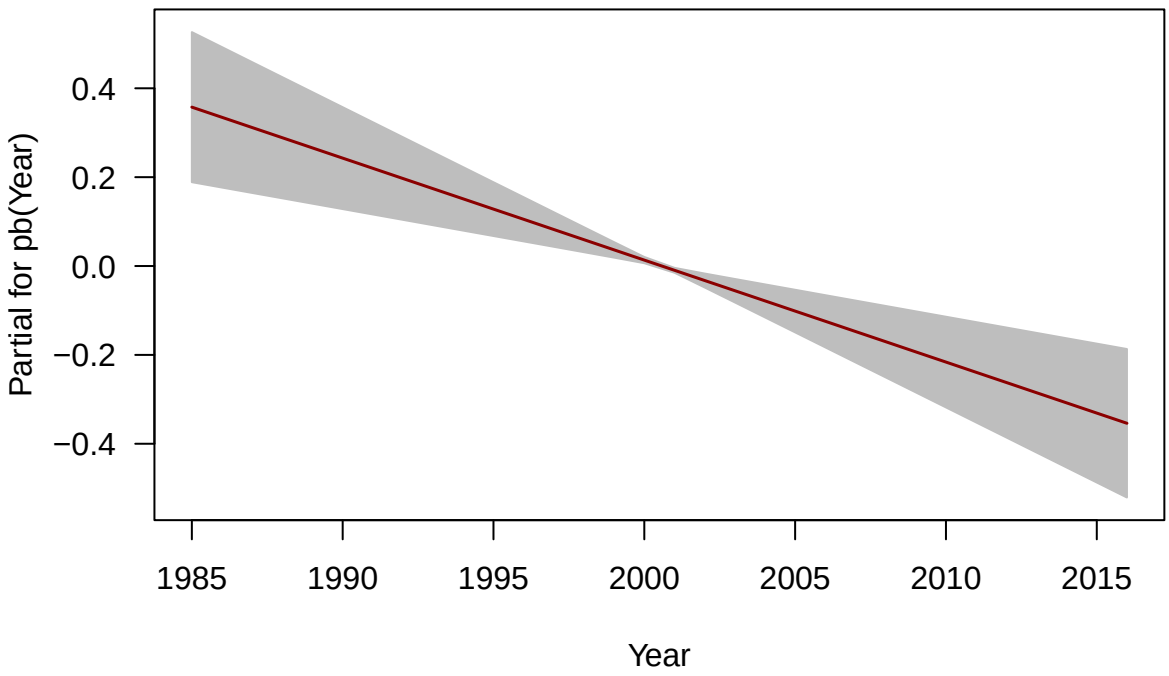

Partial for pbc(Month)
0.05
0.00
−0.05
−0.10
2
4
6
8
10
12
Month

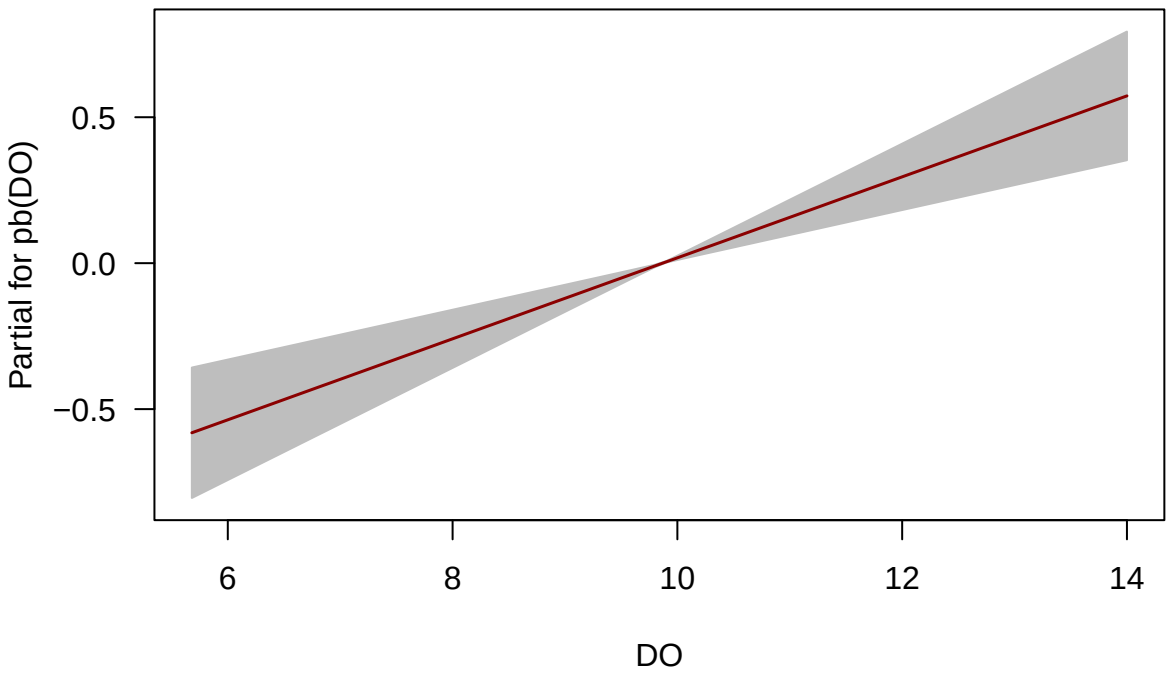

Partial for pb(SALINITY)
0.4
0.2
0.0
−0.2
5
10
15
20
SALINITY



Smooth terms for the scale (variability)

```
term.plot(m_gamlss, las = 1, se = TRUE, ylim = "free", what = "sigma")
```


Partial for pb(Year)
0.4
0.2
0.0
−0.2
−0.4
1985
1990
1995
2000
2005
2010
2015
Year



Next steps

We have considered several models without considering other variables yet. Need to carefully select the variables and a model, and an appropriate distribution family.

Variable selection in GAM is more tedious than in linear models. A general recommendation for GAM is to do forward-selection: include only relevant meaningful variables, do not just throw everything into the model. Note that the model can hold a mix of smoothed and non-smoothed terms.